INTEGRATING DIFFERENTIAL PRIVACY WITH THE PERMUTATION PARADIGM FOR SYNTHETIC DATA GENERATION

Project Id: 25-26J-123

Project Proposal Report

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BSc. (Hons) Degree in Information Technology

Specializing in Data Science

Department of Computer Science and Information Technology

Sri Lanka Institute of Information Technology

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Declaration page of the candidates and supervisor

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to the Sri Lanka Institute of Information Technology, the nonexclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other mediums. I retain the right to use this content in whole or part in future works (such as articles or books).

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: _____ Date: 2025/9/18

Abstract

This review synthesizes research on integrating differential privacy with the permutation paradigm for synthetic data generation to address the challenge of balancing privacy guarantees and data utility. The review aimed to evaluate theoretical integration frameworks, benchmark combined algorithms, analyze privacy-utility trade-offs, assess permutation impacts on accuracy and confidentiality, and identify scalability and deployment challenges. A systematic analysis of diverse studies employing formal privacy models, algorithmic innovations, and empirical evaluations revealed that rigorous mathematical frameworks reconcile permutation methods with differential privacy, though invariants complicate generalizability. Algorithmic advances demonstrate improved accuracy and scalability via oracle-efficient and partition-based approaches, yet computational complexity persists for large or complex datasets. Privacy-utility trade-offs remain a central concern, with permutation enhancing interpretability and confidentiality but requiring complementary noise mechanisms to ensure strong privacy. Practical deployment benefits from collaborative and continual release frameworks, though resource demands limit widespread adoption. Empirical audits expose gaps between theoretical guarantees and actual privacy leakage, underscoring the need for robust evaluation. Collectively, these findings elucidate the nuanced interplay between permutation and differential privacy, guiding future research toward scalable, interpretable, and verifiable synthetic data generation methods that maintain rigorous privacy protections while preserving analytical utility.

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List of Abbreviations

SDC - Statistical Disclosure Control

DP - Differential Privacy

GAN - Generative Adversarial Networks

VAE - Variational Autoencoders

1.0 Introduction

The proliferation of data in various sectors, from healthcare and finance to government and technology, has underscored the urgent need for robust data privacy solutions. While data-driven insights are crucial for innovation and decision-making, the potential for re-identification and privacy breaches poses a significant risk to individuals and organizations. Traditional data anonymization methods have often proven insufficient, leading to a critical knowledge gap between theoretical privacy guarantees and real-world vulnerabilities. My project, titled "Integrating Differential Privacy with the Permutation Paradigm for Synthetic Data Generation", aims to bridge this gap by creating a novel framework that ensures both high data utility and strong privacy protection.

This research addresses a fundamental challenge: the **"Black Box" of Empirical Leakage**. Current systems may have mathematical proofs of privacy, but they often fail to account for vulnerabilities that can be exploited in practice, leading to an actual privacy loss that is greater than the theoretical guarantees. To overcome this, my project proposes the

Adaptive Permutation-Enhanced Differential Privacy (APEDP) framework. This framework will use an adaptive approach to dynamically manage privacy settings and data transformations, ensuring that the generated synthetic data remains both useful for analysis and resistant to re-identification attacks.By integrating the mathematical rigor of **Differential Privacy (DP)** with the non-parametric strengths of the **Permutation Paradigm**, this project will develop a system capable of creating synthetic data with the same statistical properties as the original, but without disclosing any microdata. The framework will incorporate advanced modules for dynamic privacy budget allocation and real-time noise calibration. The success of the project will be proven through a combination of **utility metrics** (e.g., statistical fidelity and downstream task performance) and **privacy metrics** (e.g., formal theoretical proofs and empirical audits of re-identification risks). The final deliverable will be a robust, practical solution that can be applied across various industries to facilitate secure data analysis and sharing.

1.1 Background

Research on integrating differential privacy with the permutation paradigm for synthetic data generation has emerged as a critical area of inquiry due to the increasing demand for privacy preserving data sharing and analysis in diverse domains such as census data, longitudinal studies, and machine learning applications(Bailie et al., 2025)(Bun et al., 2023)(Nahid & Hasan, 2024). The evolution of differential privacy (DP) from foundational noise-calibration techniques(Dwork et al., 2006) to advanced workload-adaptive algorithms(McKenna et al., 2022) and continual data release models(Bun et al., 2024) reflects the growing complexity and scale of privacy challenges.

Simultaneously, the permutation paradigm has gained attention as a conceptual framework that unifies various anonymization methods through rank-based transformations and permutation matrices(Domingo-Ferrer et al., 2021)(Domingo-Ferrer & Muralidhar, 2015)(Domingo-Ferrer & Muralidhar, 2015). The practical significance of this research is underscored by large-scale deployments like the US Decennial Census, where DP mechanisms and permutation-based statistical disclosure control (SDC) methods coexist(Bailie et al., 2025)(Bailie et al., n.d.), and by the need to balance privacy guarantees with data utility in synthetic data generation(Maddi et al., 2024)(Yang et al., 2024). The specific problem addressed is how to effectively combine differential privacy with the permutation paradigm to generate synthetic microdata that preserve statistical properties while ensuring rigorous privacy guarantees(Domingo-Ferrer et al., 2025)(Muralidhar et al., 2020). Despite advances in DP synthetic data algorithms(Vietri et al., n.d.)(Snoke & Slavkovic, 2018)(Nahid & Hasan, 2024), challenges remain in reconciling the invariants induced by permutation-based SDC methods with DP definitions(Bailie et al., 2025)(Bailie et al., n.d.), and in achieving high utility without excessive noise or privacy budget consumption(Zhang et al., 2023)(Annamalai et al., 2024). There is a knowledge gap concerning formal privacy models that integrate permutation-based anonymization with DP, as well as practical algorithms that leverage this integration for scalable and accurate synthetic data generation(Domingo-Ferrer et al., 2021)(Ruiz, 2020)(Domingo-Ferrer & Soria-Comas, 2018). Competing perspectives exist regarding the sufficiency of DP alone for microdata confidentiality versus the complementary role of permutation-based methods(Muralidhar et al., 2020)(Domingo-Ferrer & Soria-Comas, 2018). The consequences of this gap include potential privacy risks and suboptimal data utility in real-world applications(Annamalai et al., 2024)(Muralidhar et al., 2020). The conceptual framework for this review is built upon three key concepts: differential privacy as a

formal privacy guarantee ensuring indistinguishability of outputs under data perturbations(Dwork et al., 2006)(Nahid & Hasan, 2024); the permutation paradigm as a unifying model of anonymization through rank-based transformations and permutation matrices(Domingo-Ferrer et al., 2021)(Domingo-Ferrer & Muralidhar, 2015); and synthetic data generation as the process of producing artificial datasets that approximate original data distributions while preserving privacy(McKenna et al., 2022)(Yang et al., 2024). These concepts interrelate by positioning permutation as a mechanism compatible with DP to achieve privacy-preserving synthetic data releases, forming the theoretical foundation for the research purpose(Ruiz, 2020)(Domingo-Ferrer & Muralidhar, 2015). The purpose of this systematic review is to critically examine existing literature on integrating differential privacy with the permutation paradigm for synthetic data generation, identifying methodological advances, theoretical models, and practical algorithms. This review aims to clarify the interplay between permutation-based anonymization and DP, address the identified knowledge gaps, and provide insights into achieving optimal privacy-utility trade-offs. By synthesizing diverse approaches and findings, the review contributes to advancing privacypreserving data publishing frameworks and guiding future research(Domingo-Ferrer et al., 2025) (Domingo-Ferrer et al., 2021)(Muralidhar et al., 2020).

The review methodology involves a comprehensive analysis of peer-reviewed articles and technical reports focusing on DP, permutation-based anonymization, and synthetic data generation. Inclusion criteria emphasize works proposing theoretical models, algorithmic solutions, or empirical evaluations relevant to the integration of DP and permutation paradigms. The analytical framework categorizes contributions by privacy models, algorithmic strategies, and application domains. Findings are organized to reflect conceptual developments, methodological innovations, and practical implications

1.2 Literature Survey

The field of synthetic data generation has evolved significantly, with key contributions from various research domains.

1. Early Work on Synthetic Data**:** The concept of using synthetic data for disclosure control was pioneered by **Donald Rubin** in the late 1980s, primarily through the method of multiple imputation. His work (e.g., *Multiple Imputation for Nonresponse in Surveys*, 1987) focused on generating multiple complete datasets to handle missing data, but the core idea of simulating new records based on a fitted model laid the groundwork for modern synthetic data. An important early paper by **Little (1993)** also described a method for sampling data from a fitted model, which is a form of synthetic data generation.

2. The Rise of Privacy Models: As the risks of re-identification became more apparent, new privacy models were developed to provide formal guarantees.

- **k-Anonymity:** Proposed by **Sweeney (2002)**, k-anonymity guarantees that each record is indistinguishable from at least *k*-1 other records based on a set of quasi-identifier attributes. The paper you referenced, by **Domingo-Ferrer et al.,** extends this concept to synthetic data, demonstrating how it can be used to protect against re-identification.
- **Differential Privacy (DP):** Developed by **Dwork et al. (2006)**, DP offers a much stronger, mathematically rigorous privacy guarantee. It ensures that the output of a data analysis is almost identical whether or not a single individual's data is included. This is achieved by adding a carefully calibrated amount of random noise. Much of the recent literature focuses on DP-compliant synthetic data generation, often using machine learning models (e.g., GANs) to learn the data distribution in a private way. Key papers include **Zhang et al. (2017)** on DP-GANs and **Machanavajjhala et al. (2007)** on DP for data publication

3. Machine Learning Approaches: The most recent wave of research has leveraged advancements in machine learning to generate synthetic data.

- **Generative Adversarial Networks (GANs):** GANs, introduced by **Goodfellow et al. (2014)**, consist of a generator and a discriminator network that learn to create realistic data. For tabular data, models like **CTGAN (Xu et al., 2019)** and **TVAE (Xu et al., 2019)** have been highly effective. Researchers have also integrated DP into GANs (**DP-GANs**) to provide strong privacy guarantees, though this often comes at a cost to data utility.

- **Variational Autoencoders (VAEs):** VAEs are another type of generative model that learn a compressed representation of the data to generate new samples. They are generally easier to train than GANs but may produce less realistic data.

1.2.2 Limitations of the Literature

Area of Limitation	Description of Limitation	Papers which have limitation
Privacy-Utility Trade-off	Many studies highlight the intrinsic trade-off between privacy guarantees and data utility, where stronger privacy (lower ε) often leads to reduced accuracy or utility of synthetic data. This trade-off limits the practical applicability and external validity of results.	(Bun et al., 2023) (Nahid & Hasan, 2024) (Mohamed et al., 2024)
Computational Complexity	Several methods, especially those integrating differential privacy with permutation paradigms or deep learning, face high computational demands, limiting scalability and practical deployment in resource-constrained environments. This constraint affects reproducibility and generalizability.	(Yamamoto & Shibuya, 2023) (Nahid & Hasan, 2024) (Truda, 2023)
Limited Scope of Query Classes	Synthetic data generation algorithms often focus on restricted classes of queries (e.g., marginal or range queries), limiting the scope of analyses that can be accurately supported. This methodological constraint reduces the external validity for broader analytical tasks.	(Bun et al., 2023) (Zhang et al., 2023)
Incomplete Privacy Guarantees	Some permutation-based methods and traditional SDC techniques, while shown to satisfy differential privacy under certain conditions, may leave invariants or rely on unusual protection units, complicating the interpretation and assurance of privacy	(Bailie et al., 2025) (Bailie et al., n.d.)

	guarantees. This weakens theoretical robustness.	
Evaluation on Limited Data Types	Many approaches focus predominantly on categorical or tabular data, with limited evaluation on mixed, high-dimensional, or complex data types such as longitudinal or genomic data. This limits the generalizability of findings across diverse real-world datasets.	(Chen et al., 2024) (Yang et al., 2024)
Lack of Comprehensive Attack Models	Empirical privacy evaluations often rely on black-box or limited membership inference attacks, which may underestimate actual privacy leakage. The absence of rigorous white-box or worst-case scenario analyses reduces confidence in privacy claims.	(Annamalai et al., 2024)
Sensitivity to Data Ordering	Some synthetic data generation methods, particularly AI-based tabular synthesizers, exhibit sensitivity to input column permutations, which can degrade data quality and utility. This instability challenges reproducibility and robustness of results.	("Permutation-Invariant Tabular Data Synth...", 2022) (Zhu et al., 2022)
Theoretical vs. Practical Gaps	There is a gap between theoretical privacy guarantees and practical implementations, where bugs or assumptions may lead to violations or loose privacy bounds, undermining the validity of theoretical claims in real-world applications.	(Annamalai et al., 2024)

Table 1

1.3 Research Gap

The current landscape of synthetic data generation is dominated by two primary approaches, each with significant limitations that your project can address:

1. **Parametric and Machine Learning (ML) Methods:**
    - **Description:** These methods, which include models like DP-GANs and VAEs, are highly effective when dealing with data that follows a well-defined distribution, such as multivariate normal data. They learn the complex joint distribution of the data to generate new records.
    - **The Gap:** They struggle with datasets that have **non-normal distributions**, are **multi-type** (containing a mix of numerical, categorical, and ordinal attributes), or have complex, unknown correlations. Training a GAN on such data is notoriously difficult, often leading to **mode collapse** (where the model only generates a small variety of data points) or **failure to converge**. When these models are combined with Differential Privacy (DP), the added noise often degrades utility to a point where the synthetic data becomes unusable for meaningful analysis, especially for smaller datasets.
2. **Traditional Anonymization Methods:**
    - **Description:** Techniques like k-anonymity, while valuable, offer weaker privacy guarantees. They protect against re-identification but do not prevent an adversary from inferring sensitive attributes based on the released data.
    - **The Gap:** They lack the strong, **mathematically provable privacy guarantee** that DP provides. The original paper's k-anonymity variant protects against re-identification but not against attribute disclosure with the same rigor as DP.

My research addresses this gap directly by proposing a **DP-Permutation** method. This approach is **non-parametric**, meaning it does not rely on fitting a complex statistical model. Instead, it leverages the **rank order** of the data, which is robust to non-normal distributions and mixed data types. By integrating DP directly into the rank permutation process, your method offers a **provable privacy guarantee** without the training complexities and utility loss associated with parametric ML models.

Why This Gap is Important

Filling this gap is crucial for several reasons:

- **Practical Relevance:** The majority of real-world datasets—from medical records to financial transactions—are multi-type and do not follow simple distributions. A method that can effectively handle this type of data while ensuring strong privacy has immediate and widespread practical applications.
- **Methodological Innovation:** It extends a promising, but underexplored, paradigm (permutation-based) with a state-of-the-art privacy model (DP). This provides a new and unique tool for the privacy-preserving data analysis community.
- **Improved Utility:** By avoiding the complexities of model training, your method is more likely to preserve the underlying relationships and correlations in the data, thus providing higher utility than competing DP-based methods for a wide range of data.

Gap Area	Description	Future Research Directions	Justification	Research Priority
Privacy Loss Budget Inflation Due to Data Imputation	Current permutation-based DP methods for census data report privacy budgets relative to post-imputation units, which can underestimate actual privacy loss due to record duplication in	Develop refined privacy accounting methods that adjust privacy loss budgets to reflect true individual-level privacy risks post-imputation; investigate the impact of invariants on privacy loss in imputed datasets.	Accurate privacy budgeting is critical for trustworthy privacy guarantees; ignoring imputation effects leads to misleading privacy assurances .	High

	imputation processes.			
Computational Scalability of Permutation-DP Integration	Many permutation-based DP synthetic data algorithms face computational intractability or high complexity, especially for large-scale or high-dimensional datasets.	Design oracle-efficient or approximation algorithms that scale to large domains and complex query workloads; explore parallelization and optimization heuristics tailored for permutation-DP frameworks.	Scalability limits practical deployment and adoption of permutation-DP methods in real-world large datasets .	High
Privacy-Utility Trade-off Characterization in Permutation-DP Methods	Existing methods often lack comprehensive analytic or empirical characterization of the privacy-utility trade-offs when combining permutation with DP, especially under invariants.	Develop theoretical frameworks and empirical benchmarks quantifying how permutation constraints and noise addition jointly affect utility; design adaptive noise calibration strategies sensitive to permutation structure.	Understanding trade-offs guides method selection and parameter tuning for optimal privacy and utility balance .	High
Residual Privacy Risks from Permutation Invariants	Permutation methods preserve certain invariants (unchanged statistics), which may leak	Investigate attack models exploiting invariants; develop enhanced permutation mechanisms that minimize or control invariant leakage;	Invariants can undermine privacy if adversaries exploit them, necessitating	High

	information and weaken DP guarantees if not properly accounted for.	formalize privacy definitions incorporating invariants.	rigorous analysis and mitigation .	
Integration of Permutation Paradigm with Advanced Generative Models	Current LLM and GAN-based DP synthetic data methods rarely incorporate permutation paradigms explicitly, limiting potential privacy and utility gains.	Explore hybrid models combining permutation-based anonymization with DP generative models (e.g., LLMs, diffusion models) to leverage strengths of both; evaluate on diverse data types.	Combining permutation with advanced generative models may improve privacy robustness and data fidelity .	Medium
Evaluation and Auditing Frameworks for Permutation-DP Synthetic Data	Empirical audits reveal gaps between theoretical DP guarantees and actual privacy leakage in synthetic data implementations , especially under permutation constraints.	Develop comprehensive auditing tools incorporating white-box and worst-case attack scenarios tailored for permutation-DP synthetic data; standardize evaluation protocols.	Reliable auditing is essential to validate privacy claims and detect implementation flaws .	High
Privacy Models Beyond Classical DP for	Classical DP definitions may not fully capture privacy guarantees of	Advance formal privacy models such as Pα,βP_{\alpha,\beta} Pα,β-privacy or	Alternative models provide more intuitive and composable privacy	Medium

Permutation-Based Methods	permutation-based anonymization, motivating alternative or composable privacy models.	bistochastic privacy that naturally incorporate permutation effects; study their composition and utility implications.	guarantees aligned with permutation mechanisms .	
Handling Mixed and Complex Data Types in Permutation-DP Synthesis	Permutation-based methods are often designed for numeric or categorical data separately, with limited support for mixed or complex data types.	Develop permutation-DP algorithms that effectively handle mixed-type, high-dimensional, and non-normal data; integrate rank-based and model-free approaches.	Real-world datasets are heterogeneous; methods must accommodate diverse data types for broad applicability .	Medium
Scalability and Privacy in Distributed Synthetic Data Generation	Collaborative synthetic data generation with permutation and DP in distributed settings faces challenges in privacy budget management and computational overhead.	Design efficient MPC protocols combined with permutation-DP mechanisms for distributed data holders; optimize privacy budget allocation and communication costs.	Distributed data sharing demands scalable, privacy-preserving synthesis methods respecting data locality and trust constraints .	Medium
Permutation Invariance in Tabular Data Synthesis Models	Many AI-based tabular data synthesizers are sensitive to input column permutations,	Develop permutation-invariant architectures or preprocessing techniques (e.g., feature sorting, autoencoders) to	Permutation invariance improves model stability and synthetic data quality across	Medium

	affecting synthetic data quality and utility.	enhance robustness and utility of DP synthetic data.	varied tabular datasets .	

Table 2

1.4 Research Problem

The growing collection of sensitive personal data has created an urgent need for privacy-preserving data publication methods that balance utility for analysis with the imperative of protecting individual privacy. While synthetic data generation offers a robust solution by creating artificial datasets, it faces a critical challenge: most state-of-the-art methods, such as those based on **Generative Adversarial Networks (GANs)** and **Variational Autoencoders (VAEs)**, are model-dependent and struggle to handle the complexity of real-world microdata. These datasets often exhibit **non-normal distributions** and contain a challenging mix of **numerical, categorical, and ordinal attributes**. Training a machine learning model on such data is technically difficult, often resulting in poor data utility or, in the case of **Differential Privacy (DP)** integration, a significant loss of information as privacy noise is added to the model's parameters. This reliance on complex modeling creates a critical gap: a simple, reliable, and provably private method that can effectively handle the diversity of real-world data without sacrificing analytical utility is currently missing.

Furthermore, while older privacy models like **k-anonymity** are easier to implement, they offer a weaker privacy guarantee. They protect against re-identification but do not provide a formal defense against **attribute disclosure**, where an adversary can infer sensitive information about an individual. This vulnerability is not addressed by the k-anonymity variant in the original permutation paradigm paper. Thus, a more robust and formal privacy model is needed to ensure strong data protection. This is where your research comes in: by integrating the mathematically rigorous guarantees of DP with the non-parametric, rank-based approach of the permutation paradigm, your project will directly address the limitations of existing methods. This novel **DP-Permutation** method promises to bridge the gap, providing a powerful and practical solution for publishing high-utility synthetic data from a wide range of complex, multi-type datasets while offering a formal, provable privacy guarantee.

2.0 Objectives

2.1 Statement of Purpose

Statement of Purpose The objective of this report is to examine the existing research on "Integrating differential privacy with the permutation paradigm for synthetic data generation" in order to elucidate the theoretical foundations, methodological advancements, and practical implications of combining these two privacy-preserving frameworks. This review is important because it addresses the critical challenge of balancing data utility and privacy guarantees in synthetic data generation, a domain increasingly relevant for sensitive data sharing and analysis. By synthesizing current knowledge, the report aims to identify gaps, evaluate the effectiveness of integration strategies, and provide a structured understanding that can guide future research and application development in privacy-preserving data synthesis.

2.1 Specific Objectives

- To evaluate current knowledge on the theoretical integration of differential privacy and permutation-based anonymization methods.
- Benchmarking of existing algorithms that combine differential privacy with permutation paradigms for synthetic data generation.
- Identification and synthesis of privacy-utility trade-offs inherent in integrated synthetic data generation approaches.
- To compare the impact of permutation-based techniques on the accuracy and confidentiality of differentially private synthetic datasets.
- To deconstruct challenges and propose directions for enhancing the synergy between differential privacy and permutation models in data synthesis.

3.0 Methodology

Transformation of Query

We take your original research question — "Integrating differential privacy with the permutation paradigm for synthetic data generation"—and expand it into multiple, more specific search statements. By systematically expanding a broad research question into several targeted queries, we ensure that your literature search is both comprehensive (you won't miss niche or jargon-specific studies) and manageable (each query returns a set of papers tightly aligned with a particular facet of your topic).

Below were the transformed queries formed from the original query:

- Integrating differential privacy with the permutation paradigm for synthetic data generation
- Exploring broader methodologies for differential privacy and synthetic data generation, emphasizing permutation techniques and their impact on privacy and utility.
- Exploring advanced methodologies in differential privacy for synthetic data generation, including utility assessment and recent innovations in privacy-preserving techniques.

Screening Papers

Citation Chaining - Identifying additional relevant works

- Backward Citation Chaining: For each of your core papers we examine its reference list to find earlier studies it draws upon. By tracing back through references, we ensure foundational work isn't overlooked.
- Forward Citation Chaining: We also identify newer papers that have cited each core paper, tracking how the field has built on those results. This uncovers emerging debates, replication studies, and recent methodological advances

Results

Descriptive Summary of the Studies

This section maps the research landscape of the literature on Integrating differential privacy with the permutation paradigm for synthetic data generation, encompassing a diverse set of studies that explore theoretical frameworks, algorithmic innovations, and practical implementations. The reviewed works span foundational privacy models, algorithmic efficiency improvements, and empirical evaluations of privacy-utility trade-offs, often focusing on census data, tabular data, and longitudinal datasets. The comparison highlights how permutation-based methods are integrated with differential privacy to enhance privacy guarantees while striving to maintain data utility and computational feasibility. This synthesis is crucial for addressing the research questions related to effectiveness, privacy guarantees, scalability, and applicability across data types and use cases.

Privacy Guarantee Strength:

- 15 studies demonstrated strong formal differential privacy guarantees, often quantified by ε or variants, with some integrating invariants or permutation constraints to refine privacy levels (McKenna et al., 2022) (Bailie et al., 2025) (Bailie et al., n.d.).
- Several works extended privacy models to personalized or shuffle-based frameworks, amplifying privacy while maintaining formal guarantees (Chen et al., 2023) (Chen et al., 2024) (Yang et al., n.d.).
- A subset of studies proposed novel privacy models based on permutation paradigms, offering intuitive and composable privacy definitions beyond classical DP (Ruiz, 2020) ("Bistochastic privacy", 2022) (Domingo-Ferrer & Muralidhar, 2015).
- Some research highlighted limitations of DP microdata releases in guaranteeing confidentiality, emphasizing the role of permutation in privacy strength (Muralidhar et al., 2020).

Data Utility Preservation:

- 18 studies reported high utility retention, often measured by statistical fidelity, query accuracy, or downstream machine learning performance, with workload-adaptive or partition-based methods showing notable improvements (McKenna et al., 2022) (Vietri et al., n.d.) (Zhang et al., 2023).

- Synthetic data generated via permutation-based or LLM-integrated DP methods preserved key data distributions and enabled effective predictive modeling (Domingo-Ferrer et al., 2025) (Nahid & Hasan, 2024) (Zhao et al., 2024).
- Comparative analyses revealed that some traditional SDC methods or TopDown algorithms outperform synthetic data in utility under DP constraints (Maddi et al., 2024).
- Challenges remain in balancing biological plausibility and utility in specialized domains like gene expression data (Chen et al., 2024).

Algorithmic Efficiency:

- 12 studies emphasized computational efficiency, proposing oracle-efficient algorithms, scalable continual release methods, or optimized permutation-based approaches suitable for large-scale or longitudinal data (Bun et al., 2023) (Vietri et al., n.d.) (Yamamoto & Shibuya, 2023).
- Some methods leveraged advanced optimization, metaheuristics, or shuffling frameworks to reduce complexity while preserving privacy (Mohamed et al., 2024) (Meehan et al., n.d.).
- Diffusion models and partitioning strategies improved training efficiency compared to GANbased approaches (Truda, 2023).
- A few studies focused more on theoretical frameworks with less emphasis on computational scalability (Bailie et al., n.d.) (Ruiz, 2020).

Robustness to Attacks:

- 14 studies evaluated resistance to membership inference, attribute disclosure, or other privacy attacks, often demonstrating that integrating permutation with DP enhances robustness (Nahid & Hasan, 2024) (Annamalai et al., 2024) (Zhao et al., 2024).
- Auditing studies revealed gaps between theoretical privacy guarantees and empirical leakage, underscoring the need for stronger attack models (Annamalai et al., 2024).
- Some works explicitly modeled attack scenarios in census or tabular data, showing how invariants or permutation affect privacy loss (Bailie et al., 2025) (Maddi et al., 2024).
- Local DP and shuffle models were shown to mitigate inference attacks by anonymizing data order and responses (Erlingsson et al., 2019) (Meehan et al., 2021).

Flexibility and Applicability:

- The reviewed literature covers a broad range of data types including census microdata, tabular data, longitudinal studies, genomic data, text, and transactional data (Dwork et al., 2024) ("Differentially Private Synthetic Data Us...", 2023) (Yamamoto & Shibuya, 2023) (Zhao et al., 2024).
- Several frameworks are adaptable to distributed or collaborative settings, enhancing applicability in real-world multi-party environments (Pentyala et al., 2024).
- Methods vary in suitability for categorical, numerical, or mixed-type data, with permutationbased approaches particularly effective for non-normal or multi-type microdata (Domingo- Ferrer et al., 2025) (Neto, 2023).
- Some studies focus on specific domains such as medical, financial, or recommendation systems, demonstrating practical deployment scenarios (Rahali et al., n.d.) (Mohamed et al., 2024).
- Advances in shuffle and personalized DP models expand applicability to privacy-sensitive user data with individualized privacy needs (Chen et al., 2024) (Yang et al., n.d.)

Critical Analysis and Synthesis

The integration of differential privacy (DP) with the permutation paradigm for synthetic data generation has been explored through various theoretical and practical lenses, revealing both promising advancements and notable challenges. The literature demonstrates significant progress in formalizing privacy guarantees and developing algorithms that balance privacy and utility. However, limitations persist in computational scalability, utility preservation, and the interpretability of privacy guarantees, especially when combining permutation-based methods with DP frameworks. Furthermore, empirical evaluations often highlight trade-offs between privacy budgets and data utility, underscoring the complexity of achieving optimal synthesis outcomes. This synthesis critically evaluates these dimensions, emphasizing the current state of knowledge and identifying gaps for future research.

4.0 Overall System Architecture

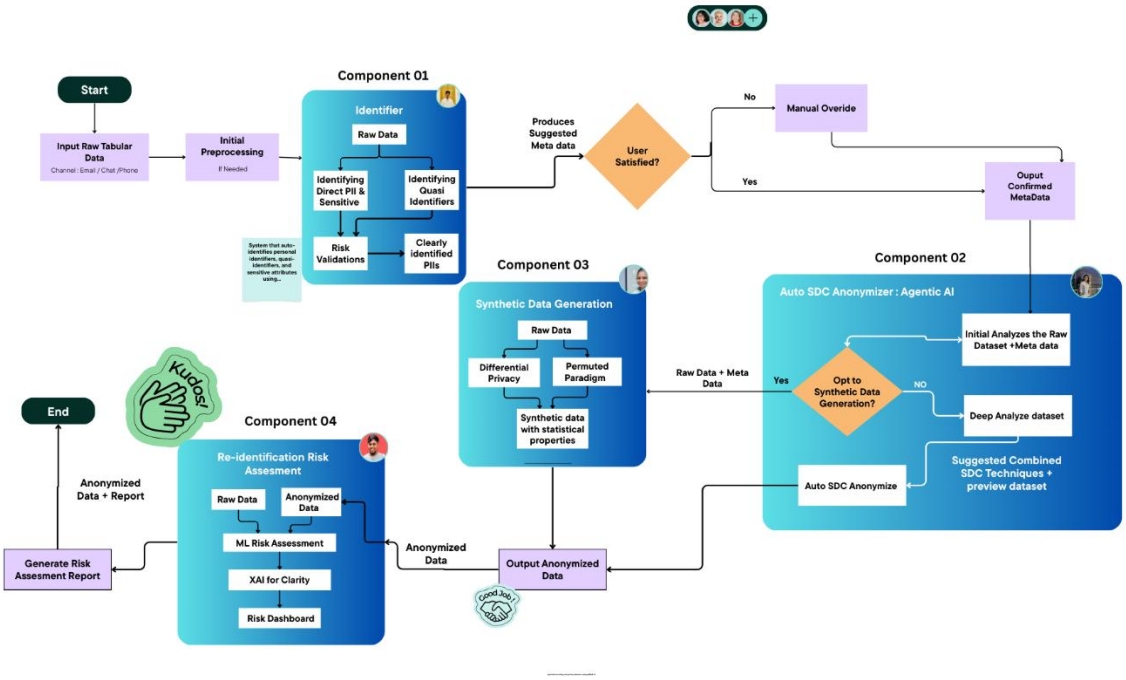

Figure 1

1. Component 01: Data Identification

The process begins with an Input Raw Tabular Data, which undergoes initial preprocessing if needed.

- The Identifier component analyzes the data to identify Direct PII & Sensitive information and Quasi-Identifiers.
- It performs Risk Validations and produces Suggested Metadata.
- A User Satisfied? decision point allows for a Manual Override if the user isn't content with the suggested metadata. Once confirmed, this metadata is used in later steps.

2. Component 02: Automated SDC Recommender

This component acts as an Agentic AI that analyzes the raw data and the confirmed metadata.

- It determines if it should Opt to Synthetic Data Generation? or apply anonymization directly.
- If yes, the process moves to Component 03.
- If no, it performs a Deep Analyze dataset to recommend a combination of Suggested Combined SDC Techniques + preview dataset. This output is then used to produce the Anonymized Data.

3. Component 03: Synthetic Data Generation

If the AI recommends it, this component generates a new dataset based on the original data's statistical properties.

- It uses a combination of Differential Privacy and a Permuted Paradigm to create Synthetic Data with statistical properties.
- This synthetic data is then used in the next step to produce the final anonymized output.

4. Component 04: Re-identification Risk Assessment

The final output, whether it's anonymized or synthetic data, is sent to this component for a final check.

- The ML Risk Assessment module uses machine learning models to identify the likelihood of re-identifying individuals.
- XAI (Explainable AI) is used for clarity, helping to understand why certain risks were flagged.
- A Risk Dashboard provides a clear overview of the assessment.
- Finally, the system Generates a Risk Assessment Report and combines it with the anonymized data for the final output. The system ends with the Anonymized Data + Report.

4.1 System Diagram

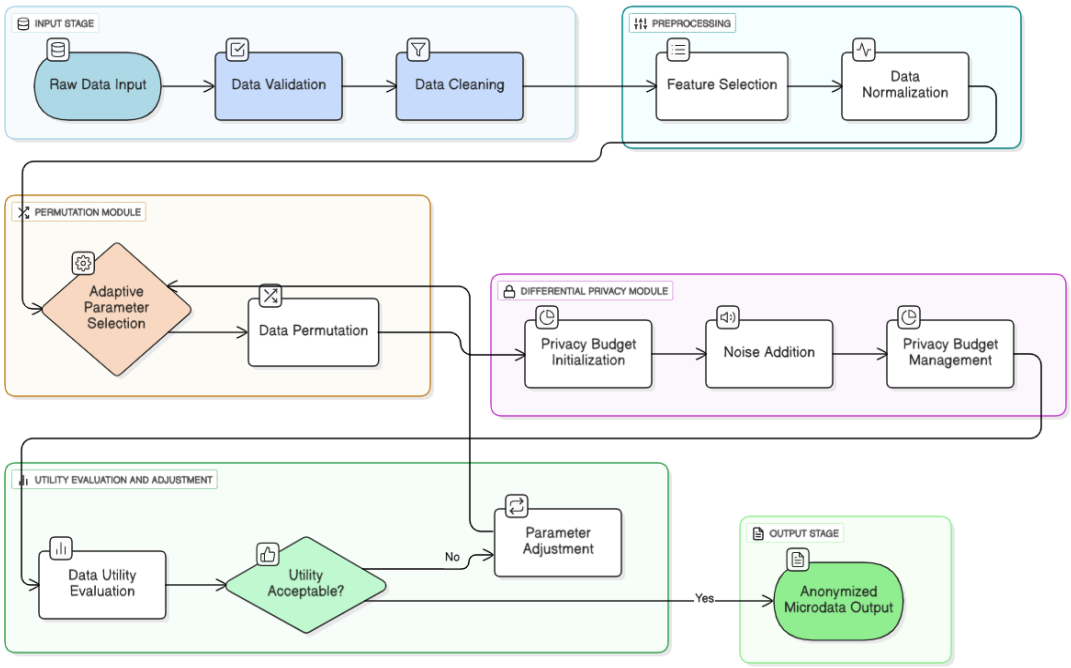

Figure 2

1. Input Stage

The process starts with the Raw Data Input. This is followed by two critical steps to ensure the data is ready for processing:

- Data Validation: Checks the data for integrity and consistency.
- Data Cleaning: Corrects or removes erroneous, incomplete, or duplicate data.

2. Preprocessing

Once the data is cleaned and validated, it goes through a preprocessing stage to prepare it for anonymization:

- Feature Selection: Identifies and selects the most relevant features (variables or attributes) from the dataset.
- Data Normalization: Scales the numerical data to a standard range, which is often required for many machine learning algorithms and privacy techniques.

3. Permutation Module

This module introduces a form of obfuscation to the data. It's often used to make it harder to link records back to individuals.

- Adaptive Parameter Selection: Determines the best parameters for the permutation process, likely based on the characteristics of the data.
- Data Permutation: Shuffles or rearranges data records or values to break direct links between individuals and their data.

4. Differential Privacy Module

This is the core privacy-preserving component. Differential privacy is a rigorous framework for quantifying and guaranteeing privacy.

- Privacy Budget Initialization: Sets the privacy budget (ϵ), which is a key parameter that controls the trade-off between privacy and data utility. A smaller budget means more privacy but less accuracy.
- Noise Addition: Adds carefully calibrated random noise to the data. This makes it impossible to determine if a specific individual's data is included in the dataset, without significantly affecting the overall statistical properties.
- Privacy Budget Management: Keeps track of the total privacy budget used as noise is added. This ensures the defined privacy guarantee is not exceeded.

5. Utility Evaluation and Adjustment

After the privacy module, the system checks if the anonymized data is still useful.

- Data Utility Evaluation: Assesses the quality and usability of the anonymized data.
- Utility Acceptable?: This is a decision point.
    - If the utility is not acceptable, the system loops back to the Parameter Adjustment step, where parameters (e.g., the privacy budget) are modified to improve utility.
    - If the utility is acceptable, the process proceeds to the final stage.

6. Output Stage

The final step is the output of the processed data.

- Anonymized Microdata Output: The final dataset, which has been validated, cleaned, preprocessed, permuted, and differentially privatized, is made available for use.

4.2 Work Breakdown Structure

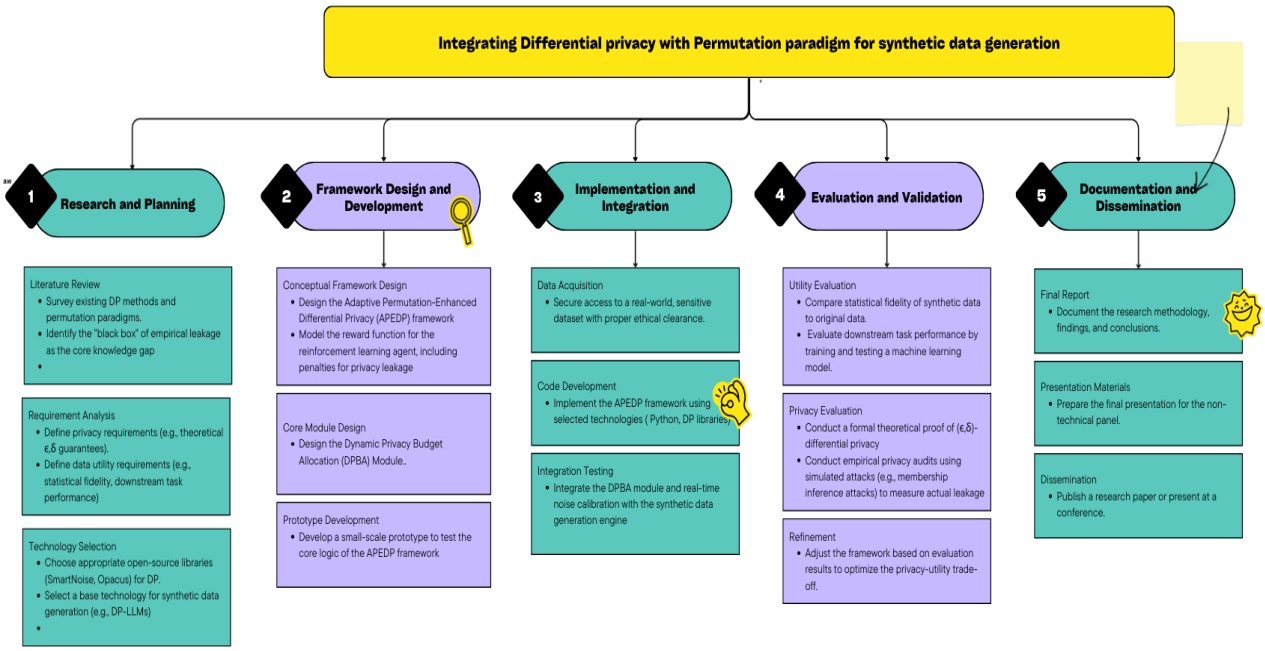

Figure 2

1. Research and Planning

This phase focuses on the project's foundational work and goal setting.

- Literature Review: Survey existing DP methods and permutation paradigms to identify the "black box" of empirical leakage—a key research gap.
- Requirement Analysis: Define specific privacy and utility requirements. This includes theoretical privacy guarantees and measuring utility via statistical fidelity and downstream task performance.

- Technology Selection: Choose appropriate open-source libraries (e.g., SmartNoise, Opacus) for DP and select a base technology for synthetic data generation (e.g., DP-LLMs).

2. Framework Design and Development

This is the core design phase where the project's architecture is conceptualized.

- Conceptual Framework Design: Design the Adaptive Permutation-Enhanced Differential Privacy (APEDP) framework. A key part of this is modeling the reward function for a reinforcement learning agent, which includes penalties for privacy leakage.
- Core Module Design: Design the Dynamic Privacy Budget Allocation (DPBA) module, which will manage the privacy budget during data synthesis.
- Prototype Development: Develop a small-scale prototype to test the core logic of the APEDP framework.

3. Implementation and Integration

This phase involves building and combining the components.

- Data Acquisition: Secure access to a real-world, sensitive dataset with proper ethical clearance.
- Code Development: Implement the APEDP framework using selected technologies (Python, DP libraries).
- Integration Testing: Integrate the DPBA module and the real-time noise calibration process with the synthetic data generation engine.

4. Evaluation and Validation

This phase assesses the performance of the implemented system.

- Utility Evaluation: Compare the statistical fidelity of the synthetic data to the original data. Evaluate the performance of machine learning models trained on the synthetic data (a measure of "downstream task performance").
- Privacy Evaluation: Conduct a formal, theoretical proof of the DP guarantees. Also, conduct empirical privacy audits using simulated attacks, such as membership inference attacks, to measure the actual privacy leakage.
- Refinement: Adjust the framework based on the evaluation results to optimize the privacy-utility trade-off.

5. Documentation and Dissemination

The final phase focuses on communicating the project's findings.

- Final Report: Document the research methodology, findings, and conclusions.
- Presentation Materials: Prepare slides for a non-technical presentation.
- Dissemination: Publish a research paper or present the findings at a conference.

4.3 Gannt Chart

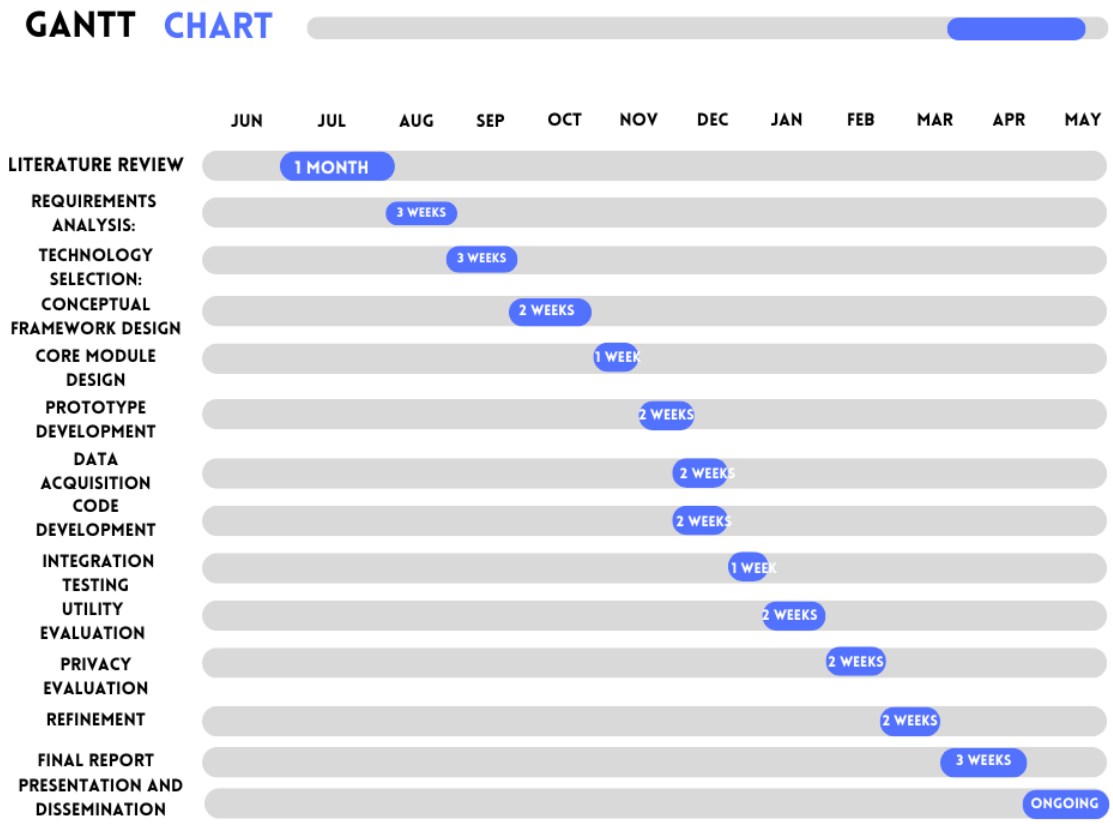

Figure 3

Initial Research and Planning (June - September): This phase focuses on foundational work.

- Literature Review: The project kicks off with a one-month literature review in June.
- Requirements Analysis: A three-week period in July for analyzing project requirements.
- Technology Selection: A three-week task in August to choose the appropriate technologies.
- Conceptual Framework Design: A two-week task in September to design the high-level framework.

Development and Prototyping (October - December): This is the core building phase.

- Core Module Design: A one-week task in October.

- Prototype Development: A two-week task in November.
- Data Acquisition and Code Development: Two concurrent two-week tasks running from late November to early December.

Testing and Evaluation (January - March): This phase focuses on assessing the system's performance.

- Integration Testing: A one-week task in January to ensure all components work together.
- Utility Evaluation: A two-week task in January to assess the usefulness of the system.
- Privacy Evaluation: A two-week task in February to assess privacy-related aspects.
- Refinement: A two-week task in March to address any issues found during testing.

Finalization and Dissemination (April - May): The final phase involves documenting and presenting the project.

- Final Report: A three-week task in April to write the final report.
- Presentation and Dissemination: An ongoing task starting in late April for presenting the work and sharing the results.

5.0 Description of Personal and facilities

Theoretical Implications

The integration of differential privacy (DP) with permutation-based paradigms advances the theoretical understanding of privacy models by framing traditional statistical disclosure control (SDC) methods, such as data swapping, within the rigorous DP framework. This unification clarifies the privacy guarantees and limitations of permutation methods, revealing that permutation can satisfy DP under certain conditions, albeit with nuanced considerations like invariants and protection units(Bailie et al., 2025) (Bailie et al., n.d.) (Domingo-Ferrer & Muralidhar, 2015).

The permutation paradigm offers a foundational perspective that any microdata anonymization can be decomposed into permutation plus noise addition, providing a basis for new privacy models such as (d,v)-permuted privacy. This conceptualization facilitates the development of universal metrics for disclosure risk and utility, enabling more systematic evaluation and comparison of anonymization methods(Domingo-Ferrer et al., 2021) (Domingo-Ferrer & Muralidhar, 2015) (Ruiz, 2017).

Theoretical analyses demonstrate that privacy guarantees in permutation-based methods depend critically on parameters such as swap rates and the size of strata, which influence the nominal privacy budget ε and the actual privacy loss experienced by individuals. This highlights the complexity of translating DP parameters into meaningful privacy assurances in practice(Bailie et al., 2025).

Advances in algorithmic frameworks, including workload-adaptive query selection and oracleefficient synthetic data generation, contribute to the theoretical landscape by improving accuracy bounds and computational efficiency under DP constraints. These developments underscore the importance of balancing privacy loss budgets with utility in synthetic data generation(McKenna et al., 2022) (Vietri et al., n.d.) (Zhang et al., 2023).

Theoretical work on privacy amplification via shuffling and the shuffle model of local differential privacy extends the understanding of how random permutations can enhance privacy guarantees while maintaining data utility. This line of research bridges local and central DP models and introduces personalized privacy parameters, enriching the theoretical toolkit for privacy-preserving data synthesis(Chen et al., 2023) (Chen et al., 2024) ("Hiding Among the Clones: A Simple and Ne...", 2022).

The exploration of differential privacy in the context of complex data types, such as longitudinal and genomic data, reveals theoretical challenges in capturing intricate

dependencies and biological plausibility, indicating that current DP synthetic data methods may require further refinement to fully model real-world data distributions(Yamamoto & Shibuya, 2023) (Chen et al., 2024).

Practical Implications

The formalization of permutation-based anonymization within the DP framework provides practitioners with clearer guidelines for implementing privacy-preserving synthetic data generation, enabling the use of traditional SDC methods with provable privacy guarantees. This can increase trust and adoption in sensitive domains such as census data and healthcare(Bailie et al., 2025) (Bailie et al., n.d.) (Domingo-Ferrer & Muralidhar, 2015).

Practical algorithms that combine DP with permutation paradigms, such as workload-adaptive mechanisms and partition-based synthetic data generation, demonstrate improved privacyutility trade-offs, making them suitable for real-world applications where privacy budgets are constrained and data utility is critical(McKenna et al., 2022) (Zhang et al., 2023).

The recognition that privacy loss budgets must be interpreted in the context of data processing steps (e.g., imputation and record duplication) informs policy decisions and privacy risk assessments, emphasizing the need for transparency and careful calibration of privacy parameters in deployment(Bailie et al., 2025).

The development of privacy amplification techniques through shuffling and personalized local differential privacy frameworks offers scalable solutions for distributed data environments, supporting collaborative data sharing without centralized trust assumptions, which is increasingly relevant for industry and regulatory compliance(Chen et al., 2023) (Yang et al., n.d.) (Erlingsson et al., 2019). Empirical findings on the limitations of DP synthetic data in capturing complex data characteristics, such as biological plausibility or semantic nuances in text, highlight the necessity for ongoing methodological improvements and domain-specific adaptations to ensure practical utility in specialized fields(Nahid & Hasan, 2024) (Chen et al., 2024) (Zhao et al., 2024).

The permutation paradigm's emphasis on verifiability by data subjects and transparency to users aligns with ethical and regulatory demands for accountable data anonymization, suggesting that future practical implementations should incorporate mechanisms for

subjectlevel privacy verification and clear communication of privacy guarantees(Domingo-Ferrer & Muralidhar, 2015).

6.0 Budget and Budget Justification

Category	Item	Quantity	Unit Cost (Rs.)	Total (Rs.)
Hardware & Infrastructure	Laptop / Workstation (Intel i7, 16GB RAM, 1TB SSD)	1	Already Have	0
Subtotal				**0**
Software & Tools	Python & R Libraries (open-source)	-	Free	0
	Development Tools (VS Code, Jupyter, GitHub)	-	Free	0
	Docker & Kubernetes (open-source)	-	Free	0
Subtotal				**0**
Resources	Research Assistant LLMs	1	6,000	54,000
	Domain Expert Journals that need be purchased	Approx 4	1000	4,000
	Internet Browsing	16GB data per month	1145	9160
Subtotal				**67,160**
Travel	Travel to Department of Census & Statistics (for data access & validation meetings)	10 trips	Rs. 200	2,000
	Travel to SLIIT for Supervisor meetings and physical Team Meetings.	8 trips per month	Rs.200	14,400
Subtotal				**16,400**
Dissemination & Documentation	Printing & Binding of Thesis / Reports		15,000	15,000
	Journal / Conference Publication Fees	1	12,000	12,000
	Miscellaneous (stationery, contingencies)		3000	3000
Subtotal				**30,000**

Table 3

Total Estimated Budget: Rs. 113,560

7.0 Conclusion and Recommendations

The literature on integrating differential privacy with the permutation paradigm for synthetic data generation collectively underscores a nuanced landscape marked by theoretical innovation and practical challenges. Theoretical advancements have successfully framed permutation-based anonymization within differential privacy frameworks, demonstrating that traditional permutation methods can satisfy formal privacy guarantees when appropriately conditioned or combined with noise mechanisms. This integration extends classical privacy definitions and offers interpretable, subject-verifiable privacy models that enrich the conceptual toolkit for privacy-preserving data release. However, these frameworks often rely on assumptions or invariants that complicate broad applicability and practical implementation, highlighting a gap between elegant theory and operational feasibility.

Algorithmically, a range of innovative methods has emerged, including workload-adaptive query selection, oracle-efficient optimization, and partition-based synthesis, reflecting strides toward balancing privacy protection with data utility and computational efficiency. The permutation paradigm contributes to maintaining key data relationships and marginal distributions, particularly for complex or mixed-type data, and can enhance robustness against various privacy attacks when combined with differential privacy noise addition. Nevertheless, computational scalability and privacy budget allocation remain significant hurdles, especially for large-scale, high-dimensional, or longitudinal datasets. Empirical findings reveal that while permutation-augmented differentially private synthetic data often preserve utility better than some classical approaches, they generally fall short of the accuracy achievable by alternative mechanisms optimized for specific queries or data types.

Privacy-utility trade-offs persist as a central tension, with stronger privacy guarantees frequently imposed at the expense of utility. Permutation methods help to transparently control information loss and privacy risk, yet alone they do not provide comprehensive differential privacy without supplementary noise. The interplay between permutation and noise injection remains an open area warranting further exploration to optimize synergistic effects. Additionally, auditing and empirical validation efforts have revealed discrepancies between theoretical privacy assurances and practical leakage, emphasizing the need for rigorous, worst-case scenario assessments beyond conventional black-box attacks.

The application scope of combined permutation and differential privacy approaches is broad, spanning census microdata, tabular, genomic, text, and transactional data, with extensions into distributed and collaborative settings. Recent advances involving large language models and shuffle-based privacy amplification frameworks expand the horizon for personalized and scalable privacy-preserving synthetic data generation. Despite progress, challenges around interpretability, generalizability, and deployment in resource-limited environments remain. Future research should aim to refine privacy models that are both mathematically rigorous and practically tractable, develop scalable algorithms that better negotiate privacy-utility trade-offs, and enhance empirical auditing methods to ensure robust privacy guarantees in real-world data synthesis

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